

# NTI Smart Home Project Report

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## Introduction

This project wants to simulate an interconnected smart home utilizing two Atmega32 microcontrollers that communicate with one another using the Universal Asynchronous Receiver-Transmitter (UART) pins found in the microcontrollers. We aim to implement our own drivers that are written in C. We are going to create the circuit and simulate the project on Proteus.

We plan on organizing the project into three main files: include, which contains the header files; lib, which contains the definitions of those files; and src, which contains two C files that are going to use the definitions we created and are then going to be compiled and loaded into the two microcontrollers. It is going to be hosted on Github, and the commits are going to be done using git commands. Our tasks are going to be distributed and marked complete in a to-do list written in our README file. Overall, the end goal of this project is to utilize the skills we have acquired this past month during our NTI embedded systems course.

## Methodology

The system was developed using a modular approach. Each hardware component was controlled through a dedicated driver to improve code organization, reusability, and maintainability. The drivers were implemented and tested individually before being integrated into the complete system.

### Drivers:

#### **DIO Driver:**

Implemented to configure and control microcontroller pins and ports. It provides the basic interface used by all other drivers for digital input and output operations.

#### **ADC Driver:**

Implemented to read analog signals from sensors and convert them into digital values. It is used for temperature measurement through the LM35 sensor and light measurement through the LDR sensor.

#### **LCD Driver:**

Implemented to control the 16×2 LCD display, including initialization, command transmission, and displaying characters and strings.

#### **Keypad Driver:**

Implemented to interface with the 4×4 keypad using the matrix scanning technique to detect user key presses.

**USART Driver:**

Implemented to provide serial communication capabilities for transmitting and receiving data between the microcontroller and external devices.

**Timer Driver:**

Implemented using Timer0 to generate timing events and keep track of passed time within the system.

**Interrupt Driver:**

Implemented to manage interrupt handling and allow the registration of custom interrupt service routines.

**Buzzer Driver:**

Implemented to control the buzzer and generate audible alerts for system notifications and warning conditions.

## Implementation:

The Smart Home System was implemented using two ATmega32 microcontrollers and several hardware and software modules to provide secure, automated, and user-friendly home control. AVR1 is responsible for the user interface and security functions, including the 4×4 keypad, LCD, password authentication, buzzer, and doorbell external interrupt. The system starts by initializing the required drivers and displaying a welcome message on the LCD. The user must enter a four-digit password through the keypad, with the entered digits displayed as asterisks for security. If the password is correct, access is granted; otherwise, the system allows the user to try again and activates the buzzer after three incorrect attempts. The LCD provides feedback about the system status, while the doorbell is handled using an external interrupt that immediately displays an alert and activates the buzzer.

AVR2 is responsible for controlling the home appliances, including the living room light, bedroom light, and fan. The system uses the ADC to read the LM35 temperature sensor and LDR light sensor, allowing automatic control based on predefined thresholds. In Automatic Mode, the fan is controlled according to the temperature, while the lighting is controlled according to the surrounding light intensity. In Manual Mode, the user can control the appliances using the keypad connected to AVR1 by sending UART commands. The DIO driver is used for digital input and output control, while the LCD, Keypad, ADC, UART, Timer, Interrupt, and Buzzer drivers provide modular control of the system components. Transistors are used as switching elements to safely control appliances that require higher current. The overall system combines security, sensor monitoring, automatic and manual appliance control, UART communication, interrupts, and timing functions to provide an integrated Smart Home solution.

Proteus:

Our project consists of two AVR's, the first microchip AVR1 works as the UI controller for our program. It also handles Security (Password) and input scanning.

**AVR1:**

- Keypad: Connected to PORTA ( PA0–PA3 for Columns, PA4–PA7 for Rows).
- LCD Display: Data lines & Control lines (RS, EN, RW) split across PORTC and PORTD. The lines RS, EN, RW are connected to PD5, PD6, PD7 and D0 to D7 are connected across all of PORTC.
- Doorbell Interruption: Push button connected to INT0 (PD2) to trigger the doorbell alarm.
- Buzzer (Alarms / Access Denied): Assigned DIO pin on PORTB pin0 .
- UART Communication (TX/RX): Connected to AVR 2 for mode status/commands.

**AVR2:**

As for the second microchip AVR2 ( Sensor & Actuator Controller), it handles the sensors (ADC), appliance Drivers, motor directional control, and system modes.

- Sensors (ADC):
  - LM35 (Temperature Sensor): Connected to PA0 (ADC0).
  - LDR (Light Sensor): Connected to PA1 (ADC1).

- Actuators / Indicators:
  - LEDs (Living Room Light, Bedroom Light, Fan Status): Connected to PORTC. The Blue LED (PC1) is used for the bedroom and the Yellow LED (PC0) for the living room .
  - DC Motor (Fan): Connected to PORTA via a BJT H-Bridge circuit (NPN 2N2222).
  
- UART Communication (TX/RX): Connected to AVR 1 for data exchange.

## Conclusion

After three solid days of work the project has finally been completed. We successfully used all the skills we've gained this past month to finish this challenging task. The efficiency we achieved through the utilization of git and Github can not be understated as it drastically improved the rate at which the project was edited by us as a team. Additionally, as a group of computer engineers learning software that simulates circuits and microcontrollers, Proteus, will be a valuable skill for the future. The mix of planning, circuitry, and programming done for this project has undoubtedly improved both our hard and soft skills.

## Data

<https://ww1.microchip.com/downloads/en/DeviceDoc/doc2503.pdf>

<https://ww1.microchip.com/downloads/en/DeviceDoc/AVR-Instruction-Set-Manual-DS40002198A.pdf>

<https://www.vishay.com/docs/37484/lcd016n002bcfhet.pdf>